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Governance Dynamics and Information Asymmetry in Decentralized Protocols

Evidence from MakerDAO/Sky Executive Spells and Voting Concentration

Facundo Villega

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Keywords

DeFi governance · GSM · MKR · voting concentration · executive spells · pre-positioning · information asymmetry

Series

Paper 3 of 4 · Villega (2026a, 2026b, 2026c, 2026d)

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Abstract (English)

We study the Governance Security Module (GSM) of MakerDAO/Sky as a dependent variable: its temporal evolution, the determinants of its changes, and the on-chain patterns of voting concentration and MKR pre-positioning preceding executive spells. The central case study is the out-of-schedule executive spell of February 20, 2025 (spell 0x1c8f5979), which reduced the GSM delay from 30 to 18 hours while simultaneously loosening five LSE-MKR-A risk parameters. H1 and H3 are presented as preliminary evidence from a single spell; their generalization to an empirical pattern requires extension to the full sample of out-of-schedule spells (§7).

First, the GSM delay series (17 changes, 2019-2025) exhibits a systematic 16h to 30h oscillation throughout 2024 (four complete cycles with constant amplitude of +/-14 hours) before the anomalous reduction to 18 hours, a value without precedent in the historical series. The subsequent corrective response (increase to 48 hours in May 2025, +30 hours) is the largest single upward adjustment since Black Thursday. The oscillation pattern is consistent with a contested equilibrium between coalitions with opposing preferences over institutional response speed, rather than a monotonic technical adjustment.

Second, we document for the February 2025 spell a structured pre-positioning sequence consistent with H1: 50,000 MKR were re-locked into the dominant voting proxy on

February 10 (ten days before cast) via a single-block consolidation through an intermediary contract, and eight wallets voted within a 52-minute window the night before execution. One wallet activated a voting proxy for the first time thirteen days before the vote, with no prior voting activity.

Third, and counterintuitively, the out-of-schedule spell exhibits lower voting concentration (HHI=3,894, 17 voters, dominant actor at 54.0%) than the immediately subsequent regular spell (HHI=9,931, dominant actor at 99.7%, effectively a single-voter outcome). This inverts the standard hypothesis that compressed deliberation windows concentrate governance power. The result is consistent with Lemma 3 of Villega (2026a): the fixed coordination cost K is consistent with low n^* in each protocol layer, but under salient out-of-schedule spells that n^* reflects explicit multi-actor coordination rather than single-actor capture. All findings are documented through 31 reproducible DuneSQL queries.

Keywords: DeFi governance · GSM · MKR · voting concentration · executive spells · pre-positioning · information asymmetry · coin voting · MakerDAO

0. Position in the Series

This paper is part of a series of four interconnected papers whose common thread is Lemma 3 of Villega (2026a): for any protocol layer in which entry requires a fixed cost K , the number of viable agents n^* is decreasing in K . The table below positions each paper in the series. The bold row corresponds to this paper.

Paper	Short Title	Layer	K interpreted as	Main Method	Dune Queries
P1	Programmatic Deposits and Monetary Transmission	Deposits (Pot/DSR)	$c_{\text{gas}} + c_{\text{audit}} + \kappa_{\text{coord}}$; $K^* \approx 2.93\text{M DAI}$	S,s model; calibration; OLS Newey-West	PIT_OU_spread_calibration_v1 (query 7384994)
P2	Auction Design and Market Concentration in Liquidations	Liquidations (Clipper/Flip)	MEV infrastructure (latency, gas, bots)	HHI; bootstrap; Mann-Whitney Monte Carlo	ethereum.logs, ethereum.transactions, ethereum.traces
P3	Governance dynamics and information asymmetry	Governance (DSChief/GSM)	Coordination cost of salient spell	31 DuneSQL queries; case analysis	p3_01–p3_31; dashboard: dune.com/facundovillega/governance-dynamics
P4	κ_{coord} as a Market Variable	Governance (κ_{coord} empirical)	$\kappa_{\text{coord_proxy}} = \text{mkr_mobilized} \times \text{gsm_delay_hours}$	OLS HC3; N=217 spells,	p4_01 (7503629), p4_02 (7486415), p4_03 (7486567)

				2019– 2025	
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1. Introduction

Mechanism design theory for decentralized protocols faces a problem that conventional governance models did not anticipate: the parameters that regulate the speed of institutional response are themselves objects of collective decision, and that decision is subject to the same information asymmetries it seeks to mitigate.

Governance power in decentralized protocols takes two empirically distinguishable forms. The first is continuous concentration: a single actor maintains dominant MKR weight across all regular spells without needing to coordinate with others. The second is episodic coordination: multiple actors mobilize within compressed windows for salient decisions. These forms have opposite implications for mechanism design: continuous concentration exposes the protocol to unilateral capture; episodic coordination indicates that critical decisions require active mobilization, implying a positive coordination cost that acts as a natural barrier. This paper shows that both forms are distinguishable on-chain and that the spell of February 20, 2025 is the case where the second form replaces the first. The analysis of the GSM delay and the February 2025 spell constitute the empirical evidence for this framework.

Table 0. Four-layer architecture — Villega Series (2026a, 2026b, 2026c, 2026d)

Layer	Object	K Mechanism	Observed n*	Paper
Layer 0	Pot (base instrument)	Not applicable	—	P1 §3.1
Layer 1	APD (sDAI, Spark)	Audit + deploy	n*=2 (86% stock)	P1 §3.2
Layer 1'	Keeper (liquidations)	MEV infrastructure	n*≈1 (97.68% Tariff)	P2 §5.4
Layer 2	Governance (MKR voters)	MKR coordination	n*=1–4 (by type)	P3 §6
Layer 3	κ_{coord} (empirical estimation)	MKR mobilized × GSM delay (217 spells, 2019–2025)	Consistent with n*=2 (consistency argument with P1)	P4 §5–6

The Governance Security Module (GSM) of MakerDAO/Sky is the clearest instance of this circularity. The GSM imposes a forced delay between the approval of an executive spell and its execution, designed to allow the market to detect and respond to adverse changes before they are implemented. But the delay itself is an adjustable parameter through a spell: any actor controlling sufficient MKR to approve a spell can modify the window that protects the protocol from that same concentration of power. This paper empirically studies that circularity.

This paper is the third of an interconnected series whose common thread is the fixed integration cost K of Lemma 3 of Villega (2026a, hereafter Paper 1). In the deposit layer, K is the deployment and audit cost of an APD, consistent with $n^*=2$ dominant agents, observed as sDAI and Spark with 86% of total stock. In the liquidation layer, K is MEV infrastructure, consistent with n^* approximately 1 under instantaneous shocks, observed as keeper 0xc721 capturing 97.68% of ETH liquidated in 67 minutes during the Tariff Shock (Villega 2026b, hereafter Paper 2). In the governance layer, K is the coordination cost to build the MKR needed to approve a salient spell, and its observed value is the object of empirical research in this paper. The empirical estimation of its dominant component — κ_{coord} — over the distribution of 217 executive spells is developed in Villega (2026d).

The GSM acts as the structural nexus of all four papers. In Villega (2026a) it is a fixed parameter that lengthens the inertia intervals of APDs: the 62% drop in spell frequency between 2020 and 2025 (Villega 2026a, Table 4.1) endogenously lengthens T_{max} , extending the periods in which APDs operate in the inertia region without governance intervention. In Villega (2026b) it is the operational classification criterion for shocks: a liquidation event that resolves before the GSM can activate is, by definition, one against which governance has no possible institutional response. The Tariff Shock, resolved in 67 minutes against a 30-hour delay, is the extreme case. In Villega (2026c), the GSM ceases to be an environmental parameter and becomes the dependent variable. In Villega (2026d), the GSM delay moves from dependent variable to primary regressor of κ_{coord} : gsm_delay_hours is the institutional driver that determines the empirically estimated coordination cost over 217 executive spells 2019-2025.

The results of Villega (2026c) feed back into Villega (2026a): if the delay is endogenously manipulable, the parameters that Villega (2026a) treats as given — spell frequency, inertia interval duration, the issuer's capacity to respond to spread deviations — are in fact variables whose stability depends on the balance of power in the governance layer documented in Villega (2026c). And they feed back into Villega (2026b): the same infrastructure for rapid coordination that allows 0xc721 to capture 97.68% of ETH in 67 minutes operates in the governance layer as the capacity for MKR pre-positioning over 10-day windows. Access asymmetry is not layer-specific; it is a structural pattern of the protocol.

2. Institutional Framework

2.1 The GSM as a Temporal Separation Mechanism

The `DSPause` (0xbe286431454714f511008713973d3b053a2d38f3) implements MakerDAO's GSM. Its function is to interpose a configurable delay between the moment a spell wins the vote in the `DSChief` and the moment it can be executed. The cycle has three stages:

Lift: The spell is proposed. In the `DSChief` (0xa618e54de493ec29432ebd2ca7f14efbf6ac17f7), it accumulates MKR until it exceeds the hat threshold. The lift event corresponds to `topic0` 0x168ccd67.

Delay: The spell remains in the queue for the configured period. It is public and inspectable: any actor can read the bytecode and evaluate its consequences.

Cast: The spell is executed (topic0 0x46d2fbbb). DSPause verifies that the spell was previously plotted, that the bytecode has not changed, and that the minimum delay time has elapsed.

The delay does not prevent governance capture: it only delays execution. Its effectiveness decreases as active participation in governance falls (Garimidi et al., 2022).

2.2 The GSM as Structural Nexus of the Series

Across the series, the GSM moves from environmental parameter in Villega (2026a), to operational classification criterion in Villega (2026b), to dependent variable here — a progression whose results feed back into the stability assumptions of the earlier papers.

Methodological note on scope: the findings of this paper are correlational. On-chain documentation establishes associations between MKR pre-positioning, GSM delay changes, and voting concentration patterns, but the specification does not identify causality in a strict sense. The analytical direction — pre-positioning anticipating an out-of-schedule spell that reduces the delay — is the natural hypothesis organizing the presentation, and is consistent with the evidence; nonetheless, causal identification would require exogenous variation in the governance calendar or in MKR availability that is not available in the single-spell sample. Findings H1-H3 are presented as preliminary evidence consistent with that hypothesis, not as causal estimation.

2.3 The Coordination Problem of Out-of-Schedule Spells

Regular spells follow an informal weekly calendar: Mondays at 14:00 UTC since 2023. An out-of-schedule spell is one executed outside that calendar, a post-hoc classification observable on the cast timestamp. The spell of February 20, 2025 was executed on a Thursday at 06:26 UTC. The distinction is relevant because out-of-schedule spells have lower anticipated public visibility and require active MKR mobilization outside the regular cycle, implying explicit coordination among holders.

3. Data

3.1 Reference Contracts

Contract	Address	Function
DSPause (GSM)	0xbe286431454714f511008713973d3b053a2d38f3	Delay between lift and cast
DSChief	0xa618e54de493ec29432ebd2ca7f14efbf6ac17f7	MKR accumulation for voting
MKR token	0x9f8f72aa9304c8b593d555f12ef6589cc3a579a2	Governance token
Feb 2025 Spell	0x1c8f5979a93ba0412677a9d315451de1570b3d03	Central case study
Main voting proxy	0x0a3f6849f78076aefadf113f5bed87720274ddc0	Custom DSProxy — not a factory vote delegate

Table 1. Reference contracts.

3.2 Data Sources

Table / Source	Content	Queries
ethereum.logs - DSPause	GSM delay history (17 events, 2019-2025)	p3_06, p3_31
ethereum.logs - DSPause	Spell inventory (lift + cast, 240 unique spells)	p3_03, p3_04
maker_ethereum.dschief_call_lock/free	Net MKR weights per voter at time of spell	p3_27, p3_28, p3_29
erc20_ethereum.evt_Transfer - MKR	MKR flows to and from voting proxies	p3_12 to p3_20
ethereum.transactions	Voter activity on proxies in prior window	p3_10, p3_11

Table 2. Data sources and associated queries.

3.3 Note on Voting Proxies

The proxy contracts identified are custom DSProxies created individually, not vote delegates created via VoteDelegateFactory. Verification was performed on maker_ethereum.votedelegatefactory_evt_createvotedelegate (query p3_26): none of the identified proxies appear in that table. This distinction is relevant: custom proxies do not expose stake metrics in MakerDAO's standard governance framework, generating an information asymmetry between those with on-chain analysis tools and those without.

3.4 spell_type Taxonomy and Precedence Rule

Priority	Type	Operational Condition
1	gsm_delay	The spell modifies the GSM delay parameter (DSPause log, queries p3_06/p3_31)
2	single_voter	Dominant actor with share $\geq 95\%$ of total MKR at time of cast
3	high_participation	total_mkr > 10,000 MKR AND dominant share < 95%
4	regular	All other cases

The precedence rule resolves overlap cases. The February 24, 2025 spell (HHI=9,931, 6 voters, dominant actor at 99.7%, total_mkr=53,441) simultaneously satisfies the conditions for single_voter (share 99.7% $\geq 95\%$) and high_participation (total_mkr=53,441 > 10,000). The hierarchy assigns single_voter because the power structure — unilateral capture — is analytically more relevant than the absolute volume of MKR. An actor with 99.7% of MKR operates without coordination constraints regardless of the total mobilized. This reclassification updates Table 8 (§6.3).

4. GSM Delay Evolution — Historical Series 2019-2025

4.1 The Complete Series

Date	Delay (h)	Delta (h)	Context
Nov 2019	0	—	Initial deploy
Feb 2020	24	+24	First configuration
Mar 2020	4	-20	Black Thursday — emergency
Apr 2020	12	+8	Post-BT: gradual recovery
Nov 2020	72	+60	Historical maximum — post-shock overprotection
Dec 2020	48	-24	Reduction to 48h
Mar 2023	16	-32	Reduction to 16h
Apr 2023	48	+32	Rise to 48h
Mar 2024	16	-32	Drop to 16h — Cycle 1
Apr 2024	30	+14	Rise to 30h — Cycle 1
Jul 2024	16	-14	Drop to 16h — Cycle 2
Aug 2024	30	+14	Rise to 30h — Cycle 2
Sep 2024	16	-14	Drop to 16h — Cycle 3
Nov 2024	30	+14	Rise to 30h — Cycle 3
Feb 2025	18	-12	Out-of-schedule spell — unprecedented value
May 2025	48	+30	Largest single increase since 2020 — corrective response
Jun 2025	24	-24	Adjustment to 24h

Table 3. Complete history of the GSM Pause Delay. Source: [ethereum.logs](#), [DSPause](#), [topic00xe177246e](#). Queries [p3_06](#) and [p3_31](#).

4.2 Three Regimes

Emergency regime (2020): Black Thursday compresses the delay from 24h to 4h in March 2020, the fastest drop in the series (-20h). Recovery is gradual: 12h in April, 72h in November (historical maximum). The delay reaches 48h in December 2020 and remains stable until 2023.

Systematic oscillation regime (2024): The delay oscillates between 16h and 30h in cycles with constant amplitude of +/-14 hours. The four semi-cycles (Mar-Apr-Jul-Aug-Sep-Nov 2024) exhibit temporal regularity without precedent in the series, consistent with a contested equilibrium between coalitions with opposing preferences.

Breakout event (Feb 2025): The February 20 spell interrupts the pattern with a reduction to 18h, a value never previously used in the series. It is neither 16h nor 30h. The subsequent reaction is the strongest since 2020: +30h in May 2025.

4.3 The 2024 Oscillation Pattern as Evidence of Contested Equilibrium

Four 16h-to-30h cycles with constant amplitude over one calendar year suggest that two coalitions with opposing preferences alternate control of the hat, a pattern cross-referenced with Chief voting weights and active votes within the prevailing delay window in §7.2. The connection to Villega (2026a) is direct: the 62% drop in spell frequency implies that the MKR needed to obtain the hat is smaller in relative terms, making the effective cost of adverse governance endogenously decreasing over time (Garimidi et al., 2022).

5. The February 20, 2025 Spell — On-chain Reconstruction

5.1 The Event and Its Parameters

Parameter	Before	After
LSE-MKR-A debt ceiling	\$20M	\$45M
LSE-MKR-A collateral ratio	200%	125%
LSE-MKR-A liquidation penalty	standard	0%
LSE-MKR-A exit fee	5%	0%
GSM Pause Delay	30 hours	18 hours

Table 4. Parameters modified by spell 0x1c8f5979 on February 20, 2025.

The simultaneous modification of risk parameters and the institutional delay in a single out-of-schedule spell is the analytically relevant element: the risk parameters directly benefit positions collateralized with MKR, while the delay reduction is associated with a shorter reaction window for potential opposing voters — an effect consistent with the hypothesis of strategic coordination but which, over a single spell, cannot be causally attributed to the delay modification without an identified counterfactual.

5.2 On-chain Timeline

Date / Time UTC	Event	Source
Feb 10, 19:45	0xb4d3be sends 20,539 MKR to 0xdeb2f1	erc20_ethereum.evt_Transfer
Feb 10, 19:49	Proxy 0x0a3f68 returns 29,062 MKR to 0xdeb2f1	erc20_ethereum.evt_Transfer
Feb 10, 20:06	0xdeb2f1 transfers 50,000 MKR to 0x3e11b0f1	erc20_ethereum.evt_Transfer
Feb 10, 20:06 (same block)	0x3e11b0f1 deposits 50,000 MKR to proxy 0x0a3f68	erc20_ethereum.evt_Transfer
Feb 5 – Feb 17	Multiple voters activate positions across 8 distinct proxies	ethereum.transactions
Feb 18, 23:34	0x37d585 votes on spell (proxy 0xde08ae)	ethereum.transactions

Feb 18, 23:36	0xfe67cb votes (proxy 0x1f68d7)	ethereum.transactions
Feb 18, 23:40	0x4c2134 votes (proxy 0xf1b837)	ethereum.transactions
Feb 18, 23:42	0xb4fb31 votes (proxy 0x0a3f68)	ethereum.transactions
Feb 18, 23:45	0xd0b444 votes (proxy 0x5d4a96)	ethereum.transactions
Feb 18, 23:56	0xd352a3 votes (proxy 0xb02e05)	ethereum.transactions
Feb 19, 00:02	0x0527c1 votes (proxy 0xe21125) — double tx	ethereum.transactions
Feb 19, 00:26	0x8b4c18 votes and schedules the spell	ethereum.transactions
Feb 20, 06:26	Cast executed by 0x8b4c18	ethereum.transactions

Table 5. On-chain timeline of the Feb 2025 spell. Queries p3_08 to p3_20.

5.3 The February 10 Re-lock Operation

In a single block on February 10, 0xdeb2f1 receives MKR from two sources, transfers 50,000 MKR to the intermediary contract 0x3e11b0f1, and that contract deposits the 50,000 MKR to proxy 0x0a3f68 in the same transaction. The proxy moves from 29,062 MKR to 50,000 MKR. The proxy was created in November 2020 and has regular institutional activity. However, 0xb4fb31e7 deposited MKR into that proxy for the first time on February 5, 2025 (13 days before the vote), with no prior deposit in any voting proxy.

5.4 The February 18 Voting Sprint

Eight distinct wallets voted the spell within 52 minutes (23:34-00:26 UTC). Selector 0xed081329 (vote(address[])) on the Chief) was called from distinct proxies with the same spell address in the data field. The last voter (0x8b4c18) voted at 00:26 and scheduled the spell 12 minutes later. The cast occurred 30 hours later — the prevailing delay that the spell itself would reduce to 18 hours upon execution.

6. Voting Concentration and Power Structure

6.1 Real MKR Weights per Voter — GSM Spell

Voter	Net MKR	Share (%)	HHI Contribution
0xdeb2f1...	20,938	54.00	2,916
0x2cba76...	9,021	23.26	541
0xdab8ae...	5,126	13.22	175
0xa53a13...	2,035	5.25	28
0xb4fb31...	605	1.56	2
Remaining (12 voters)	~1,103	2.84	~2
Total	~38,828	100	~3,894

Table 6. Net MKR weights per voter — GSM spell Feb 2025. Source: maker_ethereum.dschief_call_lock and dschief_call_free. Query p3_27.

6.2 Comparison with Regular Spell

Metric	GSM Spell (Feb 20)	Regular DSR Spell (Feb 24)
HHI	3,894	9,931
Top voter share	54.0%	99.7%
N voters with weight > 1%	4	1
Total MKR mobilized	~38,828	~53,441
Type	Out-of-schedule (Thursday 06:26)	Regular (Monday 14:00 UTC)

Table 7. Voting concentration comparison. Source: queries p3_27 and p3_28.

The out-of-schedule spell is significantly less concentrated than the regular one. In the framework of Villega (2026a) and Villega (2026b): the actor that continuously maintains MKR in the Chief does not need to coordinate with anyone (HHI=9,931, one actor at 99.7%). The out-of-schedule spell required mobilizing actors who do not normally participate — a positive and observable coordination cost (HHI=3,894, four actors with real weight).

6.3 HHI Series 2024-2025

Date	HHI	Top Share (%)	N Voters	Total MKR	Type
Mar 2024	5,126	60.4	37	7,298	regular
Mar 2024	4,232	52.7	3	106	regular
Apr 2024	1,783	26.6	19	16,759	regular
May 2024	8,957	94.6	8	2,115	regular
Jul 2024	3,322	42.6	21	2,517	regular
Aug 2024	4,632	62.9	10	4,767	regular
Oct 2024	6,806	80.1	5	76	regular
Nov 2024	8,803	93.7	15	1,709	regular
Nov 2024	9,888	99.4	8	29,227	regular
Dec 2024	4,364	60.2	6	27	regular
Jan 2025	7,625	87.1	12	318	regular
Feb 10, 2025	10,000	100.0	1	3,650	single voter
Feb 20, 2025	3,894	54.0	17	38,828	GSM delay
Feb 24, 2025	9,931	99.7	6	53,441	single_voter
Mar 2025	3,052	37.6	11	125	regular

Mar 2025	4,209	59.6	13	83	regular
Apr 2025	5,717	74.3	22	1,211	regular

Table 8. Voting HHI by spell — MakerDAO 2024-2025. Queries p3_29 and p3_30b.

The HHI exhibits no monotonic trend: it oscillates between 1,783 and 10,000. The GSM spell (HHI=3,894) is in the lower third of concentration. The February 10 spell (HHI=10,000, 1 voter, 3,650 MKR) illustrates that low participation can make the protocol vulnerable to capture with reduced capital.

The infrastructure asymmetry documented for the liquidations layer in Villega (2026b) has its direct structural analogue in this section's data. In the liquidations layer, keeper 0xc721 operated with a fleet of 19 bot-contract pairs and exclusive execution via flash loans, capturing 97.68% of ETH liquidated in 67 minutes without pre-positioned capital (Villega (2026b), §6.4). In the governance layer, proxy 0x0a3f68 consolidated 50,000 MKR in a single block on February 10 — 10 days before cast — through a three-contract chain operation, and coordinated with eight wallets in a 52-minute window the night before execution (§5.2). The mechanisms are structurally distinct: block latency speed in liquidations, capital pre-positioning in governance. But the pattern is the same: in both layers, the actor with asymmetric infrastructure captured the outcome before other participants could respond. A participant without equivalent infrastructure has no possibility of competing in the Tariff Shock or blocking the February 2025 spell, regardless of their intent or available capital.

7. Formal Hypotheses and Falsifiability Conditions

Note on scope and connection to Villega (2026d). Hypotheses H1-H3 are derived from the analysis of the February 20, 2025 spell and are presented as preliminary evidence from a single spell. H4 is an untested directional prediction in this version. The underlying question in H1-H3 — the extent to which `gsm_delay_hours` and mobilized capital determine the governance coordination cost — receives direct empirical estimation in Villega (2026d), which runs OLS HC3 over N=217 executive spells from the DSChief (2019-2025) and identifies `gsm_delay_hours` and `mkr_mobilized` as the only significant drivers at the 0.1% level ($\beta(\log \text{MKR})=1.13$; $\beta(\text{gsm_delay_hours})=0.053$; $R^2=0.672$). The reader may interpret the findings of this section as the case study motivating the specification of Villega (2026d), and the results of Villega (2026d) as the generalization of the patterns documented here over a single spell.

7.1 H1 — Structured Pre-positioning

Statement: In out-of-schedule spells, MKR accumulation in voting proxies in the 10 days before cast is structurally distinct from accumulation in regular spells. Evidence consistent with H1 is documented for the February 20, 2025 spell; its status as an empirical pattern is conditional on extension of the analysis to the full sample of out-of-schedule spells.

Preliminary evidence: The analysis documents for the February 20, 2025 spell: re-lock of 50,000 MKR 10 days before, first activation of a new wallet 13 days before, sprint of 8 voters in 52 minutes. No comparable pattern in the window prior to the regular DSR spell of February 24. Note: this finding covers a single out-of-schedule spell; its generalization

to H1 as an empirical pattern requires analysis of all identifiable out-of-schedule spells in 2020-2025 (pending extension).

Falsifiability condition: If regular spells 2024-2025 show MKR accumulation with comparable timing and magnitude, H1 must be rejected.

7.2 H2 — The Delay as an Endogenous Variable with Contested Equilibrium

Statement: The GSM delay does not follow a monotonic trajectory. The 2024 oscillation pattern reflects a contested equilibrium between coalitions with opposing preferences regarding institutional response speed.

Falsifiability condition: If voters on the 16h reductions and 30h increases are not systematically distinct, the pattern may reflect varying technical conditions rather than stable coalitions.

Evidence (H2): Testing H2 required two successive measurement strategies that yield divergent results and, in their divergence, reveal the participation structure of MakerDAO governance.

The first strategy (query H2_02) reconstructs net MKR weights deposited in the DSChief at the time of cast for the ten spells that modified the GSM delay in the 2023-2025 period. The result is negative regarding the strong form of H2: actor 0xc0583df0 appears as the first or second voter in nine of ten spells in both directions. For the first three pairs of the 2024 cycle, the overlap between the top-5 of the reduction and the immediately subsequent increase is 80-100%, consistent with the stated falsifiability condition.

The second strategy (query p3_32) restricts the universe to voters who issued a vote() call on the DSChief within the window of the prevailing delay at the time of each spell — active votes in the sole interval in which that vote could be decisive for that specific cast. Calibration of the window by spell is methodologically necessary because the delay changed ten times in the period: applying a uniform 48h window to spells with delays of 16h or 18h captures votes from the prior cycle, not from the spell in question.

The result of the calibrated window is qualitatively distinct from H2_02 and more favorable to H2: the overlap between active voters on reductions and increases is zero. Voters who issued vote() calls within the window of the prevailing delay for reductions (0x39d5, 0x29cb, 0xb9d6 in March 2023; 0xcc77, 0x0023 in February 2025) are completely disjoint from those who did so for increases (0x14a0, 0x71f1, 0x77e8 in April 2024). Actor 0xc0583df0, dominant in H2_02, does not appear in any spell with a calibrated window: its presence in H2_03 with a fixed 48h window for the April 2024 increase was an artifact — a vote cast between 16h and 48h before cast, outside the correct 16h window. Its MKR was immobile stock, not active participation in that spell.

The divergence between H2_02 and p3_32 has a structural explanation: 0xc0583df0's MKR remained continuously deposited in the Chief, without explicit re-voting per spell. Its weight in H2_02 reflects immobile stock that the Chief mechanism automatically attributes to the current slate — not necessarily an active preference over each delay change. The voters who actively drove each directional change are distinct actors. This distinction implies that the apparent dominance of a single actor in the accumulated weight metric is

partly an artifact of DSChief design, which accumulates historical MKR without requiring periodic confirmation.

The active evidence is consistent with an attenuated version of H2: not two coalitions with fixed preferences alternating the hat, but distinct activation groups that mobilize specific voters based on the direction of the proposed change.

Coverage limitation: the calibrated window only captures active voters in 3 of 10 spells. The remaining 7 do not register `vote()` calls within the prevailing delay window: either voters positioned themselves in prior cycles without explicit re-voting, or MKR remained parked from previous spells. This limitation does not invalidate the result, since the observed disjunction in spells with coverage is informative, but prevents generalizing it as a confirmed pattern across the entire series. The pending extension is to incorporate the MKR effectively active per spell via the `etch` → `slate` → `vote` chain; that chain is not resolvable in DuneSQL because the slate hash is a `keccak256` of the spell addresses array that is not invertible on-chain, which constitutes the technical limit of the analysis.

7.3 H3 — Strategic Coordination under Salient Spells (Revised Hypothesis)

Statement: Out-of-schedule spells do not exhibit greater voting concentration; they exhibit greater explicit coordination. The distinction between concentration (a single dominant actor) and coordination (multiple actors in a compressed window) is empirically observable. This hypothesis is supported by the comparison between the February 2025 GSM spell ($\text{HHI}=3,894$) and the regular February 24 spell ($\text{HHI}=9,931$); its status as a general pattern requires verification across all out-of-schedule spells in the sample.

Methodological note: This hypothesis inverts the standard prediction. If other out-of-schedule spells consistently show higher HHI than regular ones, H3 must be rejected and the standard hypothesis restored.

7.4 H4 — Surplus Buffer as Leading Signal of Adverse Governance

Statement: The state of the Surplus Buffer (weekly `flap_count` in Villega (2026a)) contains predictive information about the probability of out-of-schedule spells. This hypothesis is formulated as a directional prediction; it has not been tested in this version of the paper.

Pending evidence: Requires crossing the `flap_count` series (queries Villega (2026a): `PIT_A5_ccf_flap_dsr_v1`) with the timestamps of out-of-schedule spells. Not tested in this version; reported as a directional prediction for future work. Note: indirect evidence comes from Villega (2026a) §9.5.3, where the Surplus Buffer (`flap_count`) predicts DSR adjustments with a one-week lag; that relationship is consistent with H4 but does not test it directly.

Falsifiability condition: If `flap_count` prior to out-of-schedule spells does not differ from that prior to regular spells, H4 must be rejected.

8. Implications for Mechanism Design

8.1 The Circularity of the GSM

The central result — that the GSM delay is modifiable via the same mechanism it protects — is not an implementation flaw. It is a necessary consequence of the design. The relevant question is not how to eliminate the circularity but which parameters should be modifiable via governance and which should be immutable. Villega (2026a) (§4.3) establishes that APD exit parameters are superior in immutable form under declining participation. The same argument applies to the minimum GSM delay: a hard-coded floor would not depend on the state of governance to activate when the system needs it.

8.2 The Coordination Cost K in the Governance Layer

The February 2025 spell was approved with ~38,828 net MKR, approximately 4.3% of the circulating supply at that time. With four actors coordinating explicitly, the threshold was attainable for a small group. The 62% drop in spell frequency (Villega (2026a), Table 4.1) implies that the MKR needed to obtain the hat is smaller in relative terms as active participation falls, making the effective cost of adverse governance endogenously decreasing over time.

The February 2025 spell reduced the institutional delay from 30h to 18h while simultaneously eliminating the liquidation penalty (standard $\rightarrow$ 0%) and the exit fee (5% $\rightarrow$ 0%) of LSE-MKR-A collateral, and doubling the debt ceiling (\$20M $\rightarrow$ \$45M) with a reduced collateral ratio (200% $\rightarrow$ 125%). These changes directly benefit positions collateralized with MKR, the voters' own asset, and adversely modify the risk parameters protecting the protocol against collateral drops. The protection mechanism available to a depositor is to execute `drop()`, withdrawing funds from the Pot before the spell executes, within the delay window. With an 18h delay, that window exists in theory; in practice it requires detecting the spell within the first hours of its lock period, interpreting the implications of the five modified parameters, and executing the transaction before cast. That process requires exactly the analytical infrastructure that P3 documents as asymmetric: 31 queries running in real time over the DSPause and DSChief contracts. A Pot depositor without that infrastructure cannot execute `drop()` in time. The on-chain access asymmetry documented in §5 and §6 is not merely a transparency problem: it is the mechanism by which protocol risk parameters can be adversely modified without affected users being able to react, regardless of the level of voting concentration.

8.3 The Exposure-Control Gap: The Nexus of the Four Papers

The APDs of Villega (2026a) concentrate 86% of the deposit stock but have zero vote in the DSChief. The keepers of Villega (2026b) operate under rules established by governance but do not participate in that governance. In Villega (2026c), MKR holders who control governance have potentially divergent interests from the depositors and keepers who bear the systemic risk of their decisions. Villega (2026d) closes the cycle: it empirically estimates that the cost of that coordination (κ_{coord}) is a function of mobilized capital and the mandatory institutional delay, not of voting concentration. The same MKR voters whose coordination Villega (2026c) documents are those who generate the κ_{coord} that limits APD entry in Villega (2026a). Technical transparency of on-chain changes does not compensate for the access asymmetry to the tools needed to monitor them.

9. Limitations

First, the pre-positioning analysis covers a single out-of-schedule spell. The generalization of H1 requires analysis of all identifiable out-of-schedule spells in 2020-2025. The results of this series are derived exclusively from the MakerDAO/Sky protocol in the 2020-2026 period. The applicability of the framework to protocols with different governance architecture (Compound, Aave, Uniswap) has not been tested. In Compound, governor Bravo imposes a minimum quorum of 400,000 COMP, which alters the prediction of Lemma 3 regarding the effective cost of adverse governance. Cross-validation is a necessary extension before claiming general scope for the framework.

Second, the regular vs. out-of-schedule classification is informal, based on the observed pattern of Mondays at 14:00 UTC. Regular spells executed off-schedule for technical reasons could be incorrectly classified.

Third, voting weights are reconstructed from net deposits to the Chief within a 10-day window. This proxy understates the real weight of holders with MKR in the Chief from before the window. Table 6 should be interpreted as a lower bound for effective MKR per voter.

Fourth, H4 has not been tested. It remains a natural extension requiring time series analysis beyond the scope of this draft.

10. Conclusion

This paper documents three findings on MakerDAO/Sky governance of the GSM, articulated with the framework of Villega (2026a) and Villega (2026b) through Lemma 3 of Villega (2026a). H1 (MKR pre-positioning) and H3 (voting concentration in the out-of-schedule spell) are presented as preliminary evidence from a single spell; their generalization requires extension to the full sample of out-of-schedule spells 2020-2025.

The first finding is structural: the GSM delay is not a stable parameter but a contested variable. The 16h-to-30h oscillation pattern during 2024 and the breakout to 18h in February 2025 demonstrate that the delay reflects the balance of power between coalitions with different preferences.

The second finding is operational: the February 2025 spell was preceded by a structured, documentable on-chain pre-positioning sequence. Re-lock of 50,000 MKR ten days before, activation of a new wallet thirteen days before, sprint of eight voters in 52 minutes.

The third finding is counterintuitive and theoretically relevant: the out-of-schedule spell exhibits lower voting concentration ($HHI=3,894$) than the subsequent regular spell ($HHI=9,931$). This inverts the standard hypothesis and compels a distinction between concentration (a single dominant actor) and explicit coordination (multiple actors mobilized in a compressed window). The distinction matters because it has opposing design implications: concentration is resolved by distributing voting power; coordination is resolved by increasing coordination costs or extending the delay that makes it possible to detect.

The results directly feed back into the assumptions of Villega (2026a) and Villega (2026b): the stability of the parameters those papers treat as given depends on the balance of power in the governance layer documented in Villega (2026c).

References

- Barbon, A., Barthelemy, J. and Nguyen, B. (2025). DeFi-ying the Fed? Monetary policy transmission to stablecoin interest rates. SSRN Working Paper 4673325.
- Buterin, V. (2021). Moving beyond coin voting governance. vitalik.ca/general/2021/08/16/voting3.html
- Garimidi, P., Roughgarden, T. and Sheng, S. (2022). Decentralization of Ethereum's builder market. SSRN Working Paper.
- Harrison, J.M. and Taksar, M.I. (1983). Instantaneous control of Brownian motion. *Mathematics of Operations Research*, 8(3), 439-453.
- MakerDAO (2019). DSPause contract. Etherscan: 0xbe286431454714f511008713973d3b053a2d38f3.
- MakerDAO (2020). DSChief contract. Etherscan: 0xa618e54de493ec29432ebd2ca7f14efbf6ac17f7.
- Villega, F. (2026a). Programmatic Deposits and Monetary Transmission: A Framework of Programmatic Intermediation. Working Paper.
- Villega, F. (2026b). Auction Design and Market Concentration in MakerDAO Liquidations. Working Paper v12.
- Villega, F. (2026d). κ_{coord} as a Market Variable: Empirical Estimation over the MakerDAO DSChief, 2019-2025. Working Paper.

Appendix — Query Inventory

Range	Content
p3_01 – p3_07	DSPause event structure, spell inventory, GSM delay history
p3_08 – p3_11	Feb 2025 spell timeline, activity over voting proxies
p3_12 – p3_20	MKR flows by wallet, proxy detail, post-cast activity
p3_21 – p3_26	HHI comparison, Chief identification, vote delegate structure verification
p3_27 – p3_31	Net MKR weights, full HHI series, GSM delay with context

Appendix A. 31 reproducible queries available in [paper3_governance/queries/](#) and on the Villega (2026c) Dune dashboard.

Villega (2026c) — Working Paper v00 — In preparation. Do not cite.